

Cascaded System Performance by Model and Dataset

FP32 vs INT8 execution across ARM architectures

ASR Model	Dataset	Architecture	FP32 Cascaded Latency (ms)	INT8 Cascaded Latency (ms)	FP32 Memory Delta (MB)	INT8 Memory Delta (MB)
Whisper-base	CoVoST2	ARMv8	253,249.54	252,915.55	2.00	-0.02
Whisper-base	CoVoST2	ARMv9	50,974.32	50,860.04	2.78	1.79
Whisper-base	MSLT	ARMv8	217,027.57	216,667.26	0.72	1.22
Whisper-base	MSLT	ARMv9	49,335.28	49,214.33	2.00	1.00
Student (W1)	CoVoST2	ARMv8	220,739.91	220,441.83	0.79	0.16
Student (W1)	CoVoST2	ARMv9	51,200.78	51,105.93	2.31	-0.12
Student (W1)	MSLT	ARMv8	227,696.12	227,351.61	1.02	-0.06
Student (W1)	MSLT	ARMv9	50,126.14	49,998.59	1.54	-1.23

INT8 observation

Reported memory saving

Cascaded latency changes are small, while the table highlights the memory footprint differences across ARMv8 and ARMv9

Values are reproduced from the supplied cascaded-system performance table. Latency is total cascaded latency; memory values are reported deltas.